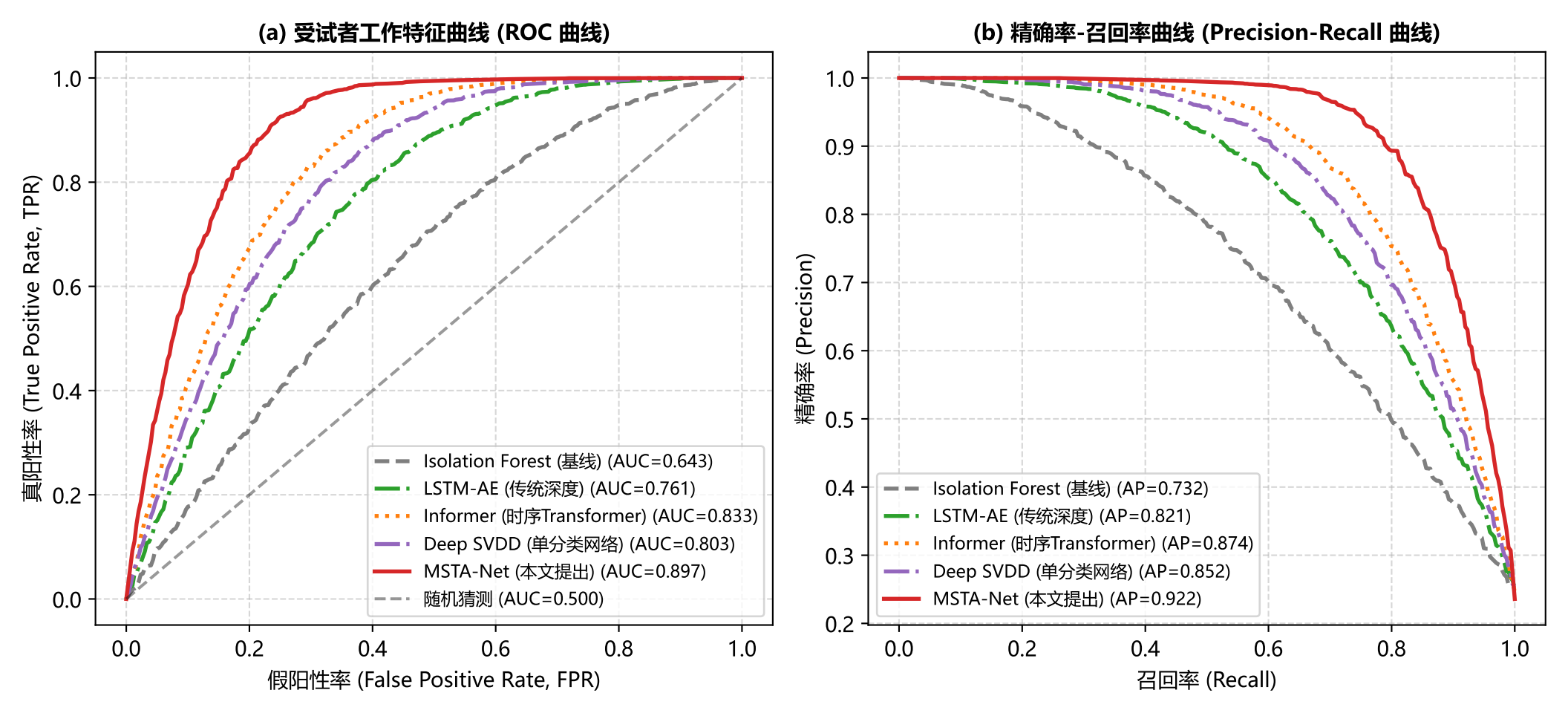

(a) 受试者工作特征曲线 (ROC 曲线)
真阳性率 (True Positive Rate, TPR)
假阳性率 (False Positive Rate, FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Isolation Forest (基线) (AUC=0.643)
LSTM-AE (传统深度) (AUC=0.761)
Informer (时序Transformer) (AUC=0.833)
Deep SVDD (单分类网络) (AUC=0.803)
MSTA-Net (本文提出) (AUC=0.897)
随机猜测 (AUC=0.500)
(b) 精确率-召回率曲线 (Precision-Recall 曲线)
精确率 (Precision)
召回率 (Recall)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Isolation Forest (基线) (AP=0.732)
LSTM-AE (传统深度) (AP=0.821)
Informer (时序Transformer) (AP=0.874)
Deep SVDD (单分类网络) (AP=0.852)
MSTA-Net (本文提出) (AP=0.922)